Chapter 5: Parametric regression models — Solutions

STA310 Survival Analysis (Master's in Applied Biostatistics)

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Chapter 5 introduces models that assume a specific probability distribution for the survival times. These exercises cover the common distributions (exponential, Weibull, log-logistic, lognormal, Gompertz), how to assess their suitability, the Weibull proportional hazards model, the accelerated failure time model, and the proportional odds model.

1 Distributions for survival times (Book §5.1)

1.1 The Weibull distribution

Type: Concept & notation

The Weibull hazard function is $h(t) = \lambda\gamma t^{\gamma-1}$ for $\lambda, \gamma > 0$.

(a) Which parameter is the *shape* parameter and which is the *scale* parameter, and how does each affect the hazard function?

(b) What special case is obtained when $\gamma = 1$?

(c) State, in words, the shape of the hazard function for $\gamma < 1$ and for $\gamma > 1$. What limitation of the Weibull hazard does this reveal?

Solution.

(a) γ is the shape parameter and λ the scale parameter. γ governs whether the hazard increases or decreases, while λ rescales the time axis.
(b) When $\gamma = 1$ the hazard is constant, $h(t) = \lambda$, and the survival times follow an exponential distribution.
(c) For $\gamma < 1$ the hazard decreases monotonically; for $\gamma > 1$ it increases monotonically. The Weibull hazard is therefore always monotonic — it cannot change direction (e.g. rise and then fall).

1.2 Hazard shapes of the common distributions

Type: Method selection

Match each situation to the most appropriate distribution, and justify briefly: exponential, Weibull, log-logistic, lognormal, Gompertz.

(a) The risk of the event is believed to be roughly constant over the follow-up period.
(b) The risk is expected to rise steadily over time (e.g. mortality increasing with age).
(c) The risk is expected to rise shortly after an intervention and then fall (e.g. death following a heart transplant, as the patient recovers).

Solution.

(a) Exponential — constant hazard, $h(t) = \lambda$.
(b) Weibull with $\gamma > 1$ (monotonically increasing), or the Gompertz distribution, whose log-hazard is linear in time and is common in demography.
(c) Log-logistic with $\kappa > 1$ (unimodal hazard), or the lognormal distribution, whose hazard rises to a maximum and then declines to zero.

2 Choosing and fitting a parametric model (Book §5.2–5.4)

2.1 The log-cumulative hazard plot

Type: Interpret figure

Builds on: Chapter 2

For a Weibull distribution, $\log\{-\log S(t)\} = \log\lambda + \gamma \log t$.

(a) How is the plot constructed, and what does a straight line indicate?
(b) What do the slope and intercept of the line estimate?
(c) You obtain a straight line with slope close to 1. What does this suggest, and what model would you then consider?

Solution.

(a) Plot $\log\{-\log \hat{S}(t)\}$ — the log-cumulative hazard, using the Kaplan–Meier estimate $\hat{S}(t)$ — against $\log t$. A straight line indicates the Weibull model is plausible.
(b) From $\log\{-\log S(t)\} = \log\lambda + \gamma \log t$, the slope estimates the shape parameter γ and the intercept estimates $\log\lambda$ (the scale parameter).
(c) A slope near 1 means $\gamma \approx 1$, so the hazard is essentially constant — the exponential distribution (a special case of the Weibull) may be adequate.

2.2 Fitting the exponential distribution

Type: Concept & notation

For a single sample from an exponential distribution with r events and total follow-up time $\sum_i t_i$, the maximum likelihood estimate is $\hat{\lambda} = r / \sum_i t_i$.

In a study of 18 women using an intrauterine device, 9 discontinued use and the total observed time was 1046 days.

(a) Compute $\hat{\lambda}$, and state the estimated hazard and survivor functions.
(b) Compute the estimated median discontinuation time, and the estimated 90th percentile.
(c) The standard error of $\hat{\lambda}$ is $\hat{\lambda}/\sqrt{r}$. Explain in words why the precision of $\hat{\lambda}$ depends on the number of events, not the number of individuals.

Solution.

(a) $\hat{\lambda} = 9/1046 = 0.0086$ per day; $\hat{h}(t) = 0.0086$ and $\hat{S}(t) = \exp(-0.0086t)$.
(b) Median $= \hat{\lambda}^{-1}\log 2 = 80.6$ days. The 90th percentile is $\hat{\lambda}^{-1}\log 10 = 267.6$ days.
(c) Each event carries the information about the hazard rate (censored times contribute exposure but no event), so the variance of $\hat{\lambda}$ is λ^2/r: it depends only on the number of events r, not on n.

2.3 Choosing between parametric and Cox models

Type: Model comparison

(a) State one advantage and one disadvantage of a Weibull proportional hazards model relative to the Cox model.
(b) When a Weibull model is a reasonable description of the data, what happens to the standard errors of the estimated coefficients compared with the Cox model?
(c) In practice, how might you decide between the two when there is little to choose in goodness of fit?

Solution.

(a) Advantage: if the distributional assumption holds, estimates are more precise (and the model yields smooth estimates of the hazard, survivor function and percentiles). Disadvantage: it imposes a specific (monotonic) form on the baseline hazard, which may be wrong; the Cox model leaves it unspecified.
(b) The standard errors are generally smaller (the model is more efficient), because it exploits the distributional assumption.

(c) Compare the standard errors of the estimated coefficients: if the Weibull model gives substantially smaller standard errors, prefer it for efficiency; if they are similar, prefer the Cox model for its less restrictive assumptions.

3 The Weibull proportional hazards model (Book §5.5–5.9)

3.1 The Weibull proportional hazards model

Type: Concept & notation

Builds on: Chapter 3

The Weibull proportional hazards model is $h_i(t) = \exp(\beta_1 x_{1i} + \cdots + \beta_p x_{pi})\, \lambda\gamma t^{\gamma-1}$.

(a) What is the baseline hazard function, and what distribution do the survival times follow?
(b) How do the explanatory variables affect the distribution of survival times — through the scale or the shape parameter?
(c) How are the β-coefficients interpreted, and what is the "proportional hazards property" of the Weibull distribution?

Solution.

(a) The baseline hazard is $\lambda\gamma t^{\gamma-1}$, the hazard of a Weibull $W(\lambda, \gamma)$ distribution — the survival times follow a Weibull distribution.
(b) They act multiplicatively on the scale parameter: the ith individual has a Weibull distribution with scale $\lambda \exp(\beta' x_i)$ and the *same* shape γ. The shape parameter is common to all individuals.
(c) Each β_j is a log hazard ratio: $\exp(\beta_j)$ is the multiplicative effect on the hazard. The Weibull has the proportional hazards property because scaling the hazard $\psi h_0(t)$ is again a Weibull hazard with the same shape γ.

3.2 Interpreting Weibull output

Type: Interpret R output

A Weibull model is fitted (in its log-linear form, $\log T_i = \mu + \alpha x_i + \sigma\epsilon_i$) to the breast cancer data, where $x = 1$ for positively stained and 0 for negatively stained tumours. The output gives $\hat{\alpha} = -0.997$ and $\hat{\sigma} = 1.067$ (reported as `Scale`).

(a) What is the estimated Weibull shape parameter γ?
(b) What does $\exp(\hat{\alpha}) = \exp(-0.997) \approx 0.37$ mean, in words?
(c) Using $\beta = -\alpha/\sigma$, compute the estimated log hazard ratio and hazard ratio for positive versus negative staining.

Solution.

(a) $\gamma = 1/\hat{\sigma} = 1/1.067 = 0.937$.
(b) It is the *time ratio*: the survival time of a positively stained woman is about 0.37 times (roughly one third) that of a negatively stained woman — positive staining accelerates death.

(c) $\hat{\beta} = -\hat{\alpha}/\hat{\sigma} = 0.997/1.067 = 0.934$, so the hazard ratio is $\exp(0.934) = 2.55$. The hazard of death for a positively stained woman is about 2.5 times that of a negatively stained woman. (This agrees with the time ratio: under Weibull the hazard ratio is $\phi^{-\gamma} = 0.37^{-0.937}$.)

4 The accelerated failure time model (Book §5.10–5.13)

4.1 The time ratio and the accelerated failure time model

Type: Concept & notation

For two groups the accelerated failure time (AFT) model is $S_N(t) = S_S(t/\phi)$.

(a) What is the parameter ϕ called, and what does $\phi < 1$ mean in terms of the time to the event?
(b) Relate ϕ to the median survival times in the two groups.
(c) If $\phi = e^{\alpha}$ and $\hat{\alpha} = -0.4$, interpret the effect of the new treatment on survival time.

Solution.

(a) ϕ is the time ratio. $\phi < 1$ means the event happens *sooner* on the new treatment (the disease process is accelerated): the survival time is a fraction ϕ of that on the standard treatment.
(b) Under the AFT model the median survival time on the new treatment is ϕ times that on the standard treatment, $t_N(50) = \phi\, t_S(50)$ — and the same holds for every percentile.
(c) $\phi = e^{-0.4} = 0.67$: the survival time on the new treatment is about 67% of that on the standard treatment, so the new treatment shortens survival (it is not beneficial).

4.2 Accelerated failure time versus proportional hazards

Type: Model comparison

(a) In words, how do the AFT and proportional hazards models differ in the way covariates act?
(b) Which single distribution possesses *both* the proportional hazards and the accelerated failure time properties, and what does this imply?
(c) In what situations might an AFT model be preferable to a proportional hazards model?

Solution.

(a) Under proportional hazards, covariates act multiplicatively on the *hazard* at each time; under AFT, covariates act multiplicatively on the *time scale*, speeding up or slowing down the whole process. A time ratio is very intuitive ("treatment doubles median survival").
(b) The Weibull distribution has both properties. As a result, the same Weibull model can be interpreted either as a PH model (via hazard ratios) or as an AFT model (via time ratios); the coefficients are related by $\beta = -\gamma\alpha$.
(c) When the proportional hazards assumption is untenable, or when the scientific question is naturally framed in terms of time (e.g. how much treatment prolongs survival), an AFT model can be more appropriate and more interpretable.

5 Other parametric models (Book §5.15–5.16)

5.1 The proportional odds model

Type: Concept & notation

In the proportional odds model, $\frac{S_i(t)}{1-S_i(t)} = e^{\eta_i}\frac{S_0(t)}{1-S_0(t)}$.

(a) What quantity do the explanatory variables act on, and how are the coefficients interpreted?
(b) What important property does the hazard ratio $h_i(t)/h_0(t)$ have in this model, and in what kind of applications is this useful?
(c) Which distribution has the proportional odds property (and also the AFT property)?

Solution.

(a) They act multiplicatively on the *odds of surviving beyond* t; each coefficient is a log odds ratio for survival beyond any time t.
(b) The hazard ratio converges from $e^{-\eta_i}$ at $t = 0$ towards unity as $t \to \infty$ — the hazards converge over time. This is useful when treatment or disease-stage effects are expected to wear off, or when a cured and a control group become more alike over time.
(c) The log-logistic distribution has both the proportional odds and the accelerated failure time properties.

6 Putting it together

6.1 A misleading interpretation

Type: What's wrong here?

A researcher fits a Weibull accelerated failure time model and obtains a time ratio of 0.5 for the new treatment. The researcher concludes: "The new treatment halves the risk of death."

Explain what is wrong with this conclusion.

Solution. The time ratio is not a hazard ratio. A time ratio of 0.5 means the survival time on the new treatment is *half* that on the standard treatment — the disease process is accelerated and the outcome is worse, not better. Under the Weibull model this corresponds to a hazard ratio of $\phi^{-\gamma} = 0.5^{-\gamma} > 1$ (for $\gamma \approx 1$, a hazard roughly *double* that of the standard). The conclusion should be that the new treatment roughly halves survival time (and thus increases the hazard), not that it halves the risk.

7 Mini-lab: fitting parametric models in R

7.1 Fitting a Weibull model with `survreg`

Type: Mini-lab

Run the code below. It simulates 300 patients in two treatment groups with Weibull survival times; the treated group has its survival times multiplied by 0.5 (an accelerated failure time effect). A Weibull model is fitted with `survreg`.

```r
set.seed(911)
n <- 300
treat <- rep(c(0, 1), each = n/2)
shape <- 0.8
scale0 <- 20
t0 <- rweibull(n/2, shape = shape, scale = scale0)
t1 <- 0.5 * rweibull(n/2, shape = shape, scale = scale0)  # time ratio 0.5
t <- c(t0, t1)
cens <- runif(n, 0, 80)
obs <- pmin(t, cens)
status <- as.numeric(t <= cens)
d <- data.frame(time = obs, status = status, treat = treat)

library(survival)
fit <- survreg(Surv(time, status) ~ treat, data = d, dist = "weibull")
summary(fit)
```

```
Call:
survreg(formula = Surv(time, status) ~ treat, data = d, dist = "weibull")
              Value Std. Error     z       p
(Intercept)  3.0312     0.1201 25.23 < 2e-16
treat       -0.6685     0.1603 -4.17 3.1e-05
Log(scale)   0.1933     0.0528  3.66 0.00026

Scale= 1.21

Weibull distribution
Loglik(model)= -857.1   Loglik(intercept only)= -865.7
    Chisq= 17.27 on 1 degrees of freedom, p= 3.2e-05
Number of Newton-Raphson Iterations: 5
n= 300
```

```r
exp(coef(fit))
```

```
(Intercept)       treat
 20.7228810   0.5124987
```

(a) What is the estimated coefficient of `treat`, and what does `exp(coef(fit))` give for `treat`? Interpret this quantity.

(b) The output reports `Scale`. What is the estimated Weibull shape parameter, and what does it say about the hazard over time?

(c) The coefficient of `treat` is on the log-time scale. What is the corresponding hazard ratio under the Weibull proportional hazards model? Is the treatment beneficial?

Solution.

(a) The estimated coefficient is about -0.67, and `exp(coef(fit))` gives about 0.51 for `treat` — the *time ratio*. The treated group's survival time is about half that of the control group (the treatment accelerates the event).

(b) `Scale` is about 1.21, so the shape parameter is $\gamma = 1/\text{Scale} \approx 0.83$ (close to the true 0.8). Since $\gamma < 1$, the hazard decreases over time.

(c) The log hazard ratio is $\beta = -\alpha/\sigma \approx 0.67/1.21 \approx 0.55$, giving a hazard ratio of about 1.7. The treatment *increases* the hazard — it is not beneficial, consistent with the time ratio below 1.